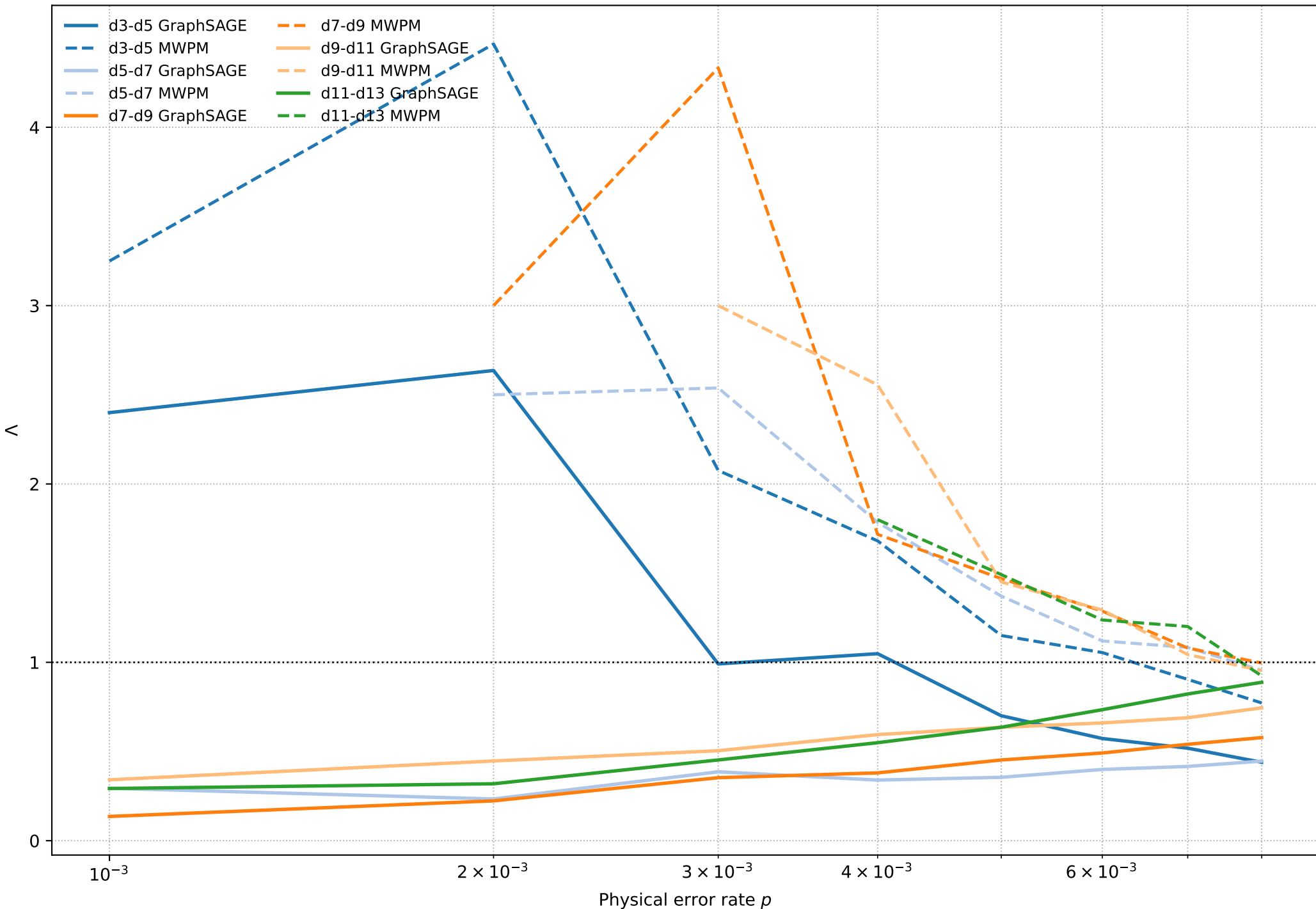


Lambda between distances ($\Lambda = p_L(d_1)/p_L(d_2)$)



GraphSAGE shows $\Lambda < 1$ across tested p ; MWPM shows $\Lambda > 1$ below threshold.
Closing the gap requires soft information + circuit-level training (see future work).